

Karan Kumar

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Professional Experience

Capgemini

Software Developer

Mumbai

Sep 2025

- Diagnose and resolve bugs in production React and Next.js applications, tracing issues across SSR rendering, API routes, and client-side state to restore functionality and improve UI reliability.
- Debug complex issues across the full stack — from frontend component behavior and API contract mismatches to backend service failures — using browser DevTools, server logs, and query execution plans to isolate root causes quickly.
- Monitor and triage data pipeline jobs in Apache Airflow and Databricks, identifying failed DAG runs, investigating root causes in transformation logic, and coordinating fixes to minimize data delivery delays.
- Write and optimize complex SQL queries against BigQuery to support data validation, pipeline debugging, and ad-hoc business reporting on large-scale datasets.
- Monitor GCP Cloud Run services and deployed environments, tracking service health, reviewing logs, and flagging anomalies to ensure uptime across production and staging.

Skills

Languages	Java, JavaScript, TypeScript, Python
Technologies	ReactJS, NodeJS, ExpressJS, NextJS, Docker, Docker compose, Kubernetes(k8s), Jenkins, Github Actions, FastAPI, Celery, Redis, WebSocket, Pytorch, LangChain, LangGraph
Database Languages	SQL, PostgreSQL, MySQL, MongoDB, TimescaleDB
Tools	Git, GitHub, Linux, VS Code, IntelliJ IDEA, Postman, Airflow, Databricks

KEY PROJECTS

Real-Time Port Logistics Intelligence System

GitHub Link

June 2026

- Architected a production-grade maritime intelligence platform ingesting live global AIS feeds, tracking **1,600+ active vessels** in real time and streaming position updates to an interactive world map via **WebSocket** — achieving sub-second latency across simultaneous connections.
- Designed a composite risk scoring engine (ETA deviation × port congestion × live weather hazards) running on a **Celery Beat scheduler** every 10 minutes, automatically classifying shipments as LOW / MEDIUM / HIGH risk and dispatching alerts to Slack and email with a configurable cooldown — reducing manual monitoring overhead to zero.
- Modeled a **TimescaleDB** time-series schema with hypertables, spatial indexes (PostGIS), and continuous aggregates for vessel positions and port congestion, enabling sub-100ms analytical queries over millions of rows without full table scans.
- Deployed a containerized **8-service stack** (FastAPI, Celery worker + beat, AIS consumer, vessel simulator, Redis, TimescaleDB, React frontend) using Docker Compose; served the frontend on **Vercel live link** and exposed the local API globally via **Cloudflare Tunnel** — full production setup with zero cloud server cost.
- Tech Stack:** Python, FastAPI, TimescaleDB, PostgreSQL, Redis, Celery, React, TypeScript, MapLibre GL, Docker, Vercel, Cloudflare

AI-Powered Cold Email Generator

GitHub Link

June 2026

- Built an AI-powered cold email generator using Python, LangChain, and Streamlit that automates personalized outreach for business development
- Integrated Groq API with Meta's **LLaMA 3.3-70B model** to extract structured job data (role, skills, experience) from scraped career pages using prompt engineering
- Implemented a **ChromaDB vector store** to semantically match job-required skills against a portfolio of 20+ tech-stack/project-link pairs using similarity search
- Developed a web scraping pipeline with **LangChain's WebBaseLoader** and regex-based text cleaning to parse any job posting URL
- Chained multiple LangChain PromptTemplates to build a two-step RAG pipeline: job extraction → context-aware email generation with relevant portfolio links
- Deployed an interactive Streamlit web UI enabling one-click cold email generation from any job URL <https://coldemailgeneratortool-4dvgypxqz5afm55xxkdozf.streamlit.app/>
- Technologies:** Python, Streamlit, LangChain, Groq API (LLaMA 3.3-70B), ChromaDB, BeautifulSoup4, Pandas

- Built a full-stack personal portfolio using **Next.js 15 App Router** with TypeScript and Tailwind CSS, deployed on Vercel with ISR (revalidate=3600) and an on-demand revalidation webhook for near-instant content updates
- Integrated **Notion as a headless CMS** for the blog — converts Notion blocks to Markdown via notion-to-md, renders with react-markdown + rehype-highlight for full syntax-highlighted code blocks
- Wired live **GitHub API** and **GeeksforGeeks API** integrations to surface real-time coding stats (repos, stars, followers, DSA problem breakdown by difficulty) using parallel server-side fetches
- Implemented UX details including **Framer Motion page transitions**, an IntersectionObserver-based sticky table of contents with active-heading tracking, and a scroll-driven reading progress bar
- Built a **contact form** backed by the Resend email API with client-side loading/success/error states, and a /api/revalidate webhook route for Notion-triggered cache purges.

KEY ACHIEVEMENTS

- Solved 400+ coding problems across CodeChef (250+) and LeetCode (150+), showcasing problem-solving skills
- Completed 100+ days of solving POTD on GeeksforGeeks, demonstrating consistent algorithmic proficiency
- Recognized by Deutsche Bank with a bag for successful completion of a competitive GeeksforGeeks coding challenge

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CCP)
- GitHub Foundation Certification
- Docker Hands-On: Containerization Fundamentals
- Kubernetes for Beginners (Udemy)
- Agile Software Development (Coursera)
- Pytorch (Udemy)

EDUCATION

Technocrats Institute of Technology

B.Tech in Computer Science (*CGPA: 7.88/10.00*)

Bhopal, MP

Oct 2021 – June 2025